3)

$\lambda^3 + 0.45\lambda^2 + 36.06\lambda + 1.802 = 0$

① ⇓ used calculator to find roots

$(\lambda + .05)(\lambda + .2 + 6j)(\lambda + .2 - 6j)$

$y_0(t) = c_1 e^{-0.05t} + c_2 e^{(-.2-6j)t} + c_3 e^{(-.2+6j)t}$

kept in this form for ease of calculating

$y_0(0) = 2 = c_1 + c_2 + c_3$

$y_0'(0) = 3 = -0.05c_1 + (-.2-6j)c_2 + (-.2+6j)c_3$

$y_0'(0) = 1 = (0.05)^2 c_1 + (-.2-6j)^2 c_2 + (-.2+6j)^2 c_3$

⇓ used calculator

$c_1 = 2.062$
$c_2 = -0.031 + .2576j$
$c_3 = -0.031 - .2576j$

$y(t) = 2.062e^{-0.05t} + (-0.031 + .2576j)e^{(-.2-6j)t} + (-0.031 - .2576j)e^{(-.2+6j)t}$

Clarification on ①

$$\lambda + .05 \;\overline{)\;\lambda^3 + .45\lambda^2 + 36.06\lambda + 1.802}\;\;\lambda^2 + .4\lambda + 36.04$$

$\lambda^3 + .05\lambda^2$

$.4\lambda^2 + 36.06\lambda$

$.4\lambda^2 + .02\lambda$

$36.04\lambda + 1.802$

$36.04\lambda + 1.802$

0

used calculator to find real roots

$(\lambda + .05)(\lambda^2 + .4\lambda + 36.04)$

$(\lambda + .05)(\lambda + .2 + 6j)(\lambda + .2 - 6j)$

used quadratic formula